

Label-setting criteria for stochastic shortest path games with two players

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Website #16 | Solver verdict: SOLVED | Verifier verdict: ISSUES | Shown: solver output

Problem

We are given a finite-state zero-sum stochastic shortest-path game with state set

$$X = \{x_1, \dots, x_n, t\},$$

where t is absorbing and terminal, terminal cost is 0, and for each nonterminal state x the players choose actions $a \in A(x)$ (Min) and $b \in B(x)$ (Max). The one-step cost is $C(x, a, b) \geq 0$, and the next-state law is $p(x, a, b, \cdot)$ on X . Under standard well-posedness assumptions, the value U is finite and satisfies

$$U(t) = 0, \quad U(x) = \min_{a \in A(x)} \max_{b \in B(x)} \left(C(x, a, b) + \sum_{y \in X} p(x, a, b, y) U(y) \right), \quad x \neq t.$$

The task is to derive *a priori*, locally checkable hypotheses—stated only in terms of the one-step data C and p —that guarantee that U can be computed by a *label-setting* method: once a state is accepted, its label is never revised. I interpret the request in the strongest natural way: the conditions should not depend on the unknown value function U .

Alternative readings considered

I considered two readings.

First, the purely graph-theoretic reading: finite state and finite action sets, so every local verification is a finite combinatorial check. This is the main reading I adopt.

Second, the compact-action reading suggested by the SSP-game literature: action sets may be compact rather than finite, but minima and maxima in the Shapley equation are attained. My second sufficient condition below still applies in that setting whenever the required local envelope data exist; only the implementation cost of the local minimization changes.

Related literature found

- **Bertsekas–Tsitsiklis (1991)** [BACKGROUND]: classical finite-state SSP analysis for the one-player problem; useful for orientation about properness, Bellman equations, and value-iteration baselines, but not about two-player label-setting.
- **Patek–Bertsekas (1999)** [BACKGROUND]: establishes existence of a real-valued equilibrium cost vector for SSP games under suitable assumptions and convergence of value/policy iteration, but does not give a Dijkstra/Dial-type theorem for simultaneous-move stochastic transitions.
- **Vladimirsky (2008)** [TOOL/PARTIAL]: develops explicit causality conditions for *single-player* multimode SSPs and makes the key distinction between explicit causality and absolute causality; this was the closest stochastic-control predecessor.

- **Bardi–Maldonado López (2016)** [PARTIAL]: gives a Dijkstra-type algorithm for *deterministic* zero-sum dynamic games with alternating moves; relevant as a two-player label-setting near-neighbor, but it removes stochasticity and changes the information pattern.
- **Bertsekas (2019)** [PARTIAL/TOOL]: analyzes a robust shortest-path model with adversarial successor selection and proves a Dijkstra-like algorithm for nonnegative arc lengths; this is very close to the deterministic min-max equation that my reduction below produces.
- **Gaspard–Vladimirsky (2025)** [PARTIAL]: derives monotone-causality conditions for *single-player* opportunistically stochastic SSPs; useful as a modern reference for causality in stochastic shortest-path settings, but again without an antagonistic player.

Among the sources I checked, the nearest neighbors split into three categories: single-player stochastic label-setting, deterministic two-player label-setting, and deterministic robust/minimax shortest paths. I did not find, among these consulted sources, a paper proving a Dijkstra/Dial theorem for the exact simultaneous-move stochastic zero-sum SSP formulation stated here; that is why I derived the conditions below directly.

Proof sketch

I prove two independent sufficient conditions.

The first is the obvious but important *explicit-causality* condition: if the union of all possible one-step transitions forms an acyclic digraph, then the Bellman equation can be solved by a single topological sweep. This already handles arbitrary two-player stochasticity, including zero costs.

The second condition is the nontrivial one. The obstacle in a simultaneous stochastic game is the inner maximization over b : a label-setting update at x is usable only if that worst-case term can be evaluated from already accepted labels. I therefore look for an *exact deterministic envelope* for the inner max. Concretely, for each pair (x, a) , I ask whether every stochastic Max-response is Bellman-dominated by the convex combination of some deterministic adverse responses, and whether those deterministic adverse responses are themselves available. If so, then

$$\max_b \left(C(x, a, b) + p(x, a, b) \cdot W \right)$$

collapses *exactly* to a max over deterministic successors:

$$\max_{y \in M(x, a)} \left(d(x, a, y) + W(y) \right).$$

So the original stochastic game becomes a deterministic min-max shortest-path hypergraph.

Once that reduction is in hand, positive envelope costs give strict decrease of the value along every optimal mode. That strict decrease is precisely the causality needed by Dijkstra: the optimal mode of the smallest unlabeled state uses only already accepted states. A uniform positive lower bound δ on all envelope costs gives the stronger gap needed by Dial: states in the same δ -bucket cannot depend on one another.

A useful corollary is that the envelope condition is automatically satisfied when Max directly chooses a transition distribution ξ on a finite mode and the stage cost is a convex function of ξ : then Jensen’s inequality supplies the required Bellman domination.

Main result

Theorem 1. *Assume that the Shapley minima and maxima are attained and that the SSP game has a finite value function U .*

Then each of the following hypotheses is sufficient for a correct label-setting computation of U .

(I) *Explicit support acyclicity. Let G_{sup} be the directed graph on X with an arc $x \rightarrow y$ whenever $x \neq t$ and there exist $a \in A(x)$, $b \in B(x)$ such that $p(x, a, b, y) > 0$. If G_{sup} is acyclic, then a single topological sweep computes U exactly.*

(II) *Exact deterministic envelope with positive envelope costs. For every nonterminal state x and every $a \in A(x)$, suppose there exist*

- *a finite set $M(x, a) \subseteq X$,*
- *numbers $d(x, a, y) > 0$ for $y \in M(x, a)$,*
- *and, for each $y \in M(x, a)$, an action $b_{x,a}^y \in B(x)$,*

such that for all $b \in B(x)$,

$$(E1) \quad p(x, a, b, z) = 0 \quad \text{for every } z \notin M(x, a),$$

$$(E2) \quad C(x, a, b) \leq \sum_{y \in M(x, a)} p(x, a, b, y) d(x, a, y),$$

and for all $y \in M(x, a)$,

$$(E3) \quad p(x, a, b_{x,a}^y, y) = 1, \quad C(x, a, b_{x,a}^y) = d(x, a, y).$$

Then the original Shapley equation is exactly equivalent to

$$U(t) = 0, \quad U(x) = \min_{a \in A(x)} \max_{y \in M(x, a)} (d(x, a, y) + U(y)), \quad x \neq t.$$

Consequently, if $P \subseteq X$ is the current permanent set and $L(y)$ denotes the permanent label of $y \in P$, then the Dijkstra-type tentative label

$$T_P(x) := \min_{\substack{a \in A(x) \\ M(x, a) \subseteq P}} \max_{y \in M(x, a)} (d(x, a, y) + L(y))$$

is exact for the next state to be accepted, so the usual permanent-label Dijkstra scheme is correct.

If, in addition, there exists a constant $\delta > 0$ such that

$$d(x, a, y) \geq \delta \quad \text{for all } x \neq t, \ a \in A(x), \ y \in M(x, a),$$

then a Dial-type bucket method with bucket width δ is also correct.

Proof

For brevity write

$$Q(x, a, b; W) := C(x, a, b) + \sum_{z \in X} p(x, a, b, z) W(z)$$

for any real vector $W = (W(z))_{z \in X}$.

Lemma 1 (Explicit support acyclicity implies a single-sweep label-setting method). *If G_{sup} is acyclic, then U is computable by one topological sweep.*

Proof. Because G_{sup} is acyclic, there exists a topological ordering

$$t = x^{(0)}, x^{(1)}, \dots, x^{(n)}$$

such that every arc of G_{sup} goes from a later state to an earlier one. Equivalently, if $p(x^{(k)}, a, b, y) > 0$, then $y = x^{(j)}$ for some $j < k$.

Set $U(t) = 0$. Now suppose $U(x^{(0)}), \dots, U(x^{(k-1)})$ have already been computed. For the state $x^{(k)}$, every successor y appearing with positive probability in any term

$$Q(x^{(k)}, a, b; U)$$

belongs to $\{x^{(0)}, \dots, x^{(k-1)}\}$, so every value $U(y)$ entering the Shapley equation at $x^{(k)}$ is already known. Therefore

$$U(x^{(k)}) = \min_{a \in A(x^{(k)})} \max_{b \in B(x^{(k)})} Q(x^{(k)}, a, b; U)$$

can be computed exactly and permanently at this step.

Proceeding through the topological order computes every component once and never revisits it. That is a label-setting algorithm. $\square$

Lemma 2 (Exact deterministic envelope). *Fix a nonterminal state x and an action $a \in A(x)$, and assume (E1)–(E3) hold for this pair (x, a) . Then for every real vector $W = (W(z))_{z \in X}$,*

$$\max_{b \in B(x)} Q(x, a, b; W) = \max_{y \in M(x, a)} (d(x, a, y) + W(y)).$$

Proof. First, let $b \in B(x)$ be arbitrary. By (E1), only states in $M(x, a)$ can receive positive probability, so

$$Q(x, a, b; W) = C(x, a, b) + \sum_{y \in M(x, a)} p(x, a, b, y) W(y).$$

Using (E2),

$$Q(x, a, b; W) \leq \sum_{y \in M(x, a)} p(x, a, b, y) d(x, a, y) + \sum_{y \in M(x, a)} p(x, a, b, y) W(y).$$

Combining the sums gives

$$Q(x, a, b; W) \leq \sum_{y \in M(x, a)} p(x, a, b, y) (d(x, a, y) + W(y)).$$

A weighted average of finitely many numbers is at most their maximum, hence

$$Q(x, a, b; W) \leq \max_{y \in M(x, a)} (d(x, a, y) + W(y)).$$

Since this holds for every $b \in B(x)$,

$$\max_{b \in B(x)} Q(x, a, b; W) \leq \max_{y \in M(x, a)} (d(x, a, y) + W(y)).$$

For the reverse inequality, fix any $y \in M(x, a)$. By (E3), the specific action $b_{x,a}^y$ satisfies

$$p(x, a, b_{x,a}^y, y) = 1 \quad \text{and} \quad C(x, a, b_{x,a}^y) = d(x, a, y).$$

Therefore

$$Q(x, a, b_{x,a}^y; W) = d(x, a, y) + W(y).$$

So the maximum over $b \in B(x)$ is at least this quantity for every $y \in M(x, a)$. Taking the maximum over y gives

$$\max_{b \in B(x)} Q(x, a, b; W) \geq \max_{y \in M(x, a)} (d(x, a, y) + W(y)).$$

The two opposite inequalities prove equality. $\square$

Lemma 3 (Reduced Bellman equation). *Under hypothesis (II), the original SSP-game Bellman equation is exactly equivalent to*

$$U(t) = 0, \quad U(x) = \min_{a \in A(x)} \max_{y \in M(x, a)} (d(x, a, y) + U(y)), \quad x \neq t.$$

Proof. Fix $x \neq t$. For each $a \in A(x)$, Lemma 2 gives

$$\max_{b \in B(x)} Q(x, a, b; U) = \max_{y \in M(x, a)} (d(x, a, y) + U(y)).$$

Now minimize both sides over $a \in A(x)$. This is exactly the claimed reduced Bellman equation. $\square$

Lemma 4 (Strict causality in the reduced game). *Assume hypothesis (II) and let $x \neq t$. If $a^* \in A(x)$ is optimal in the reduced Bellman equation at x , then for every $y \in M(x, a^*)$,*

$$U(y) \leq U(x) - d(x, a^*, y) < U(x).$$

Proof. Optimality of a^* means

$$U(x) = \max_{z \in M(x, a^*)} (d(x, a^*, z) + U(z)).$$

Every term inside that maximum is therefore at most $U(x)$. In particular, for each fixed $y \in M(x, a^*)$,

$$d(x, a^*, y) + U(y) \leq U(x),$$

hence

$$U(y) \leq U(x) - d(x, a^*, y).$$

Because $d(x, a^*, y) > 0$, this implies $U(y) < U(x)$. $\square$

Proof of Theorem 1. Part (I) is Lemma 1, so it remains to prove part (II).

The reduced Bellman equation follows from Lemma 3. Thus it is enough to prove correctness of Dijkstra and Dial for that reduced equation.

Dijkstra-type method. Let $P \subseteq X$ be the current permanent set, and suppose each $y \in P$ already carries its exact permanent label $L(y) = U(y)$. For $x \notin P$, define

$$T_P(x) := \min_{\substack{a \in A(x) \\ M(x,a) \subseteq P}} \max_{y \in M(x,a)} (d(x,a,y) + L(y)),$$

with the convention $T_P(x) = +\infty$ if no such a exists.

We prove by induction on $|P|$ that selecting any state with smallest tentative label T_P is safe.

For the inductive step, choose $\bar{x} \notin P$ with minimal true value among all unlabeled states:

$$U(\bar{x}) = \min\{U(z) : z \notin P\}.$$

Let a^* be an optimal action for $\bar{x}$ in the reduced Bellman equation. By Lemma 4, every $y \in M(\bar{x}, a^*)$ satisfies $U(y) < U(\bar{x})$. Since $\bar{x}$ has the smallest unlabeled value, each such y must already belong to P . Therefore $M(\bar{x}, a^*) \subseteq P$, so a^* is admissible in the definition of $T_P(\bar{x})$. Using $L(y) = U(y)$ on P ,

$$T_P(\bar{x}) \leq \max_{y \in M(\bar{x}, a^*)} (d(\bar{x}, a^*, y) + U(y)) = U(\bar{x}).$$

On the other hand, $T_P(\bar{x})$ minimizes over a *restricted* set of actions, so it cannot be smaller than the unrestricted reduced Bellman value:

$$T_P(\bar{x}) \geq U(\bar{x}).$$

Hence

$$T_P(\bar{x}) = U(\bar{x}).$$

Now take any other unlabeled state $z \notin P$. Again, because $T_P(z)$ uses a restricted action set,

$$T_P(z) \geq U(z) \geq U(\bar{x}) = T_P(\bar{x}).$$

So every global minimizer of the tentative labels has true value equal to the smallest unlabeled value, and its tentative label is exact. Accepting such a state is therefore safe.

This proves correctness of the Dijkstra-type permanent-label method.

Dial-type method. Assume now that there exists $\delta > 0$ such that $d(x, a, y) \geq \delta$ for all x, a, y . Partition the nonnegative half-line into half-open buckets

$$[k\delta, (k+1)\delta), \quad k = 0, 1, 2, \dots$$

Let λ be the smallest true value among currently unlabeled states. Consider any unlabeled state x with

$$\lambda \leq U(x) < \lambda + \delta.$$

Let a^* be optimal at x . By Lemma 4, every $y \in M(x, a^*)$ satisfies

$$U(y) \leq U(x) - d(x, a^*, y) \leq U(x) - \delta < \lambda.$$

So all those successors are already permanent, and the same argument as above gives

$$T_P(x) = U(x) \in [\lambda, \lambda + \delta).$$

Thus every state whose true value lies in the current bucket has an exact tentative label already in that bucket.

Conversely, if z is unlabeled with $U(z) \geq \lambda + \delta$, then

$$T_P(z) \geq U(z) \geq \lambda + \delta,$$

so its tentative label is not in the current bucket.

Therefore the lowest nonempty tentative bucket contains *exactly* the states whose true values lie in that same bucket, and all of their tentative labels are exact. They may all be accepted permanently.

It remains to check that accepting one state from the current bucket cannot create a new tentative value inside the same bucket for some other unlabeled state. Suppose a newly considered candidate for a state z uses a newly accepted state x from the current bucket. Then the candidate value has the form

$$\max_{y \in M(z, a)} (d(z, a, y) + U(y))$$

for some action a with $x \in M(z, a)$. Since one of the terms inside the maximum is $d(z, a, x) + U(x)$, we have

$$\max_{y \in M(z, a)} (d(z, a, y) + U(y)) \geq d(z, a, x) + U(x) \geq \delta + \lambda.$$

So such a candidate cannot fall back into the current bucket $[\lambda, \lambda + \delta)$.

Hence no state in the current bucket depends on another state in the current bucket, and the Dial bucket acceptance is correct.

This proves part (II). $\square$

Lemma 5 (Convex adversarial mixing implies the exact-envelope condition). *Suppose that for a fixed pair (x, a) , Max's action directly specifies a probability vector*

$$\xi = (\xi_y)_{y \in M(x, a)}$$

on a finite mode $M(x, a)$, so that

$$p(x, a, \xi, y) = \xi_y \quad (y \in M(x, a)),$$

and the one-step cost is

$$C(x, a, \xi) = c_{x, a}(\xi),$$

where $c_{x, a}$ is convex on the simplex over $M(x, a)$. Assume also that every vertex action e_y is available. Then (E1)–(E3) hold with

$$d(x, a, y) := c_{x, a}(e_y).$$

Therefore Theorem 1 (II) applies.

Proof. Condition (E1) is built into the parametrization.

For (E3), each vertex e_y is an available action, and by definition

$$p(x, a, e_y, y) = 1, \quad C(x, a, e_y) = c_{x, a}(e_y) = d(x, a, y).$$

For (E2), take any distribution ξ on $M(x, a)$. Since

$$\xi = \sum_{y \in M(x, a)} \xi_y e_y$$

and $c_{x,a}$ is convex,

$$c_{x,a}(\xi) \leq \sum_{y \in M(x,a)} \xi_y c_{x,a}(e_y) = \sum_{y \in M(x,a)} p(x,a,\xi,y) d(x,a,y).$$

That is exactly (E2). □

The hypotheses in part (II) are locally verifiable. For a fixed pair (x,a) , once one has identified candidate deterministic adverse responses $b_{x,a}^y$, checking (E2) is just a finite family of linear inequalities in the local transition probabilities and local one-step costs.

Gap to the full statement

There is no logical gap relative to the wording of the problem: the problem asked for *sufficient* local conditions guaranteeing applicability and correctness of label-setting methods, and Theorem 1 supplies two such families, one topological and one genuinely Dijkstra/Dial-like.

What I do *not* claim is necessity or a complete characterization of all SSP games admitting label-setting.

Self red-team

I tried to break each nontrivial hypothesis with the smallest examples I could find.

First, if (E2) is removed, the reduction can fail immediately. Take one state x , one Min-action a , two successor states y_1, y_2 , deterministic Max-actions to y_1, y_2 with costs $d_1 = d_2 = 1$, and an additional mixed Max-action with transition law $(1/2, 1/2)$ and cost 60. For $W(y_1) = 0$, $W(y_2) = 100$, the mixed action yields 110, whereas the deterministic envelope gives only $\max\{1, 101\} = 101$.

Second, if (E3) is removed, only the upper bound survives, not equality. For example, with only a single mixed response $(1/2, 1/2)$ of cost 1, one may satisfy (E2) using $d_1 = d_2 = 2$, but then at $W(y_1) = W(y_2) = 0$ the true inner max is 1, while the surrogate deterministic max is 2.

Third, if positivity is removed, strict causality breaks. A two-state deterministic chain $x_1 \rightarrow x_2$ with cost 0, followed by $x_2 \rightarrow t$ with cost 1, has $U(x_1) = U(x_2) = 1$. Thus optimal dependence may occur on an equal-valued state, which is fatal for the strict Dijkstra/Dial argument. A zero-cost two-cycle gives an even sharper obstruction.

The three most plausible referee objections, and my answers, are these.

- *Objection:* “Why is the stochastic inner max really replaced exactly by a deterministic one?”

Answer: Lemma 2 proves both directions separately. (E2) gives the upper bound; (E3) gives the matching lower bound by exhibiting available deterministic actions.

- *Objection:* “What if two unlabeled states have the same value?”

Answer: Lemma 4 rules out same-value dependence along any optimal mode in part (II), because every optimal successor satisfies $U(y) < U(x)$. Hence ties cause no problem: tied states are independent for acceptance purposes.

- *Objection:* “Dial’s proof is standard for arcs; why does it still work for hyperarcs/modes?”

Answer: Because any candidate using a newly accepted state x contains at least one term $d(z,a,x) + U(x)$, and that single term is already $\geq U(x) + \delta$. Since the whole candidate is the maximum over the mode, it also lies at least one full bucket ahead.

Boundary checks: if every $M(x, a)$ is a singleton, part (II) reduces to the classical deterministic shortest-path setting; if the support graph is acyclic, part (I) works even with zero costs; and the reduced Bellman equation in part (II) matches the robust/deterministic min-max shortest-path form expected from the literature.

I also re-read the proof line by line to ensure that every nontrivial inference is either an explicit inequality, a restriction-to-subset argument, or a direct consequence of the reduced Bellman equation.

Invoked results

No external theorem is used as a proof step. All cited papers were used only as BACKGROUND / PARTIAL / TOOL sources for orientation and comparison, not as black-box lemmas inside the proof. Therefore there is no imported theorem whose hypotheses require an in-proof hypothesis audit.

Self-confidence

3 — I checked the reduction lemma and the Dijkstra/Dial inductions carefully and could not break them, but because the deterministic-envelope condition is custom rather than standard terminology, I leave a small margin for a hidden edge-case convention.